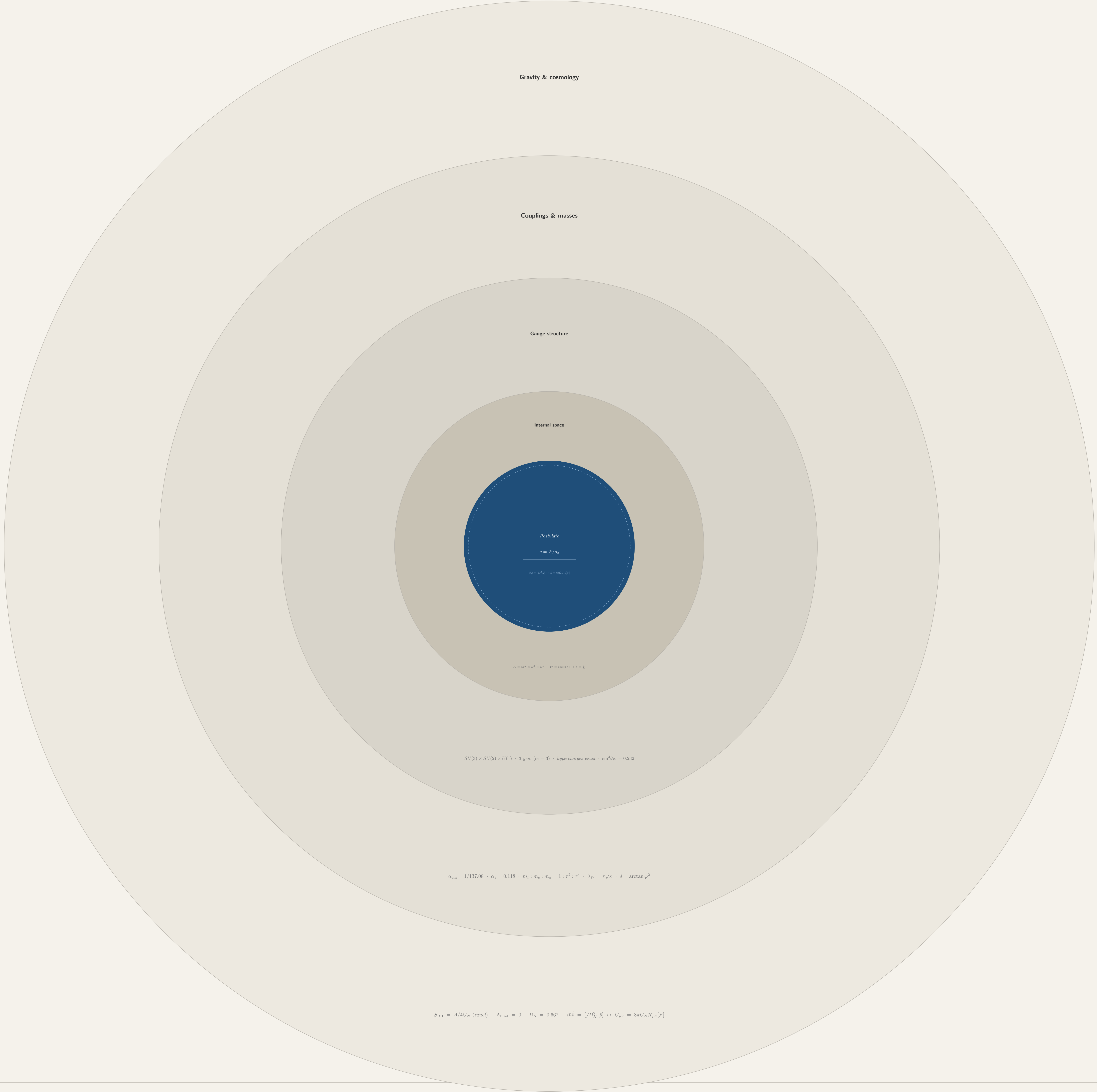




POSTULATE & UNIFIED EQUATION

$$g_{AB} = \mathcal{F}_{AB} / \rho_0$$
$$i\hbar d\rho/dt = [D_{K_1}^2, \rho] \leftrightarrow G_{\mu\nu} = 8\pi G_N \mathbb{R}_{\mu\nu}[\mathcal{F}(\rho)]$$

The spacetime metric is the Fisher information tensor. QM and GR are two aspects of one equation.

SELECTED RESULTS — ZERO FREE PARAMETERS

Weakling angle	Free parameter	Strong coupling	Hypercharges
$\sin^2 \theta_W = 0.232$ obs: 0.2312 ± 0.0015	$1/\alpha = 137.08$ obs: 137.0359 ± 0.0015	$\alpha_s = 0.118$ obs: 0.118 ± 0.001	$+\frac{1}{6}, +\frac{2}{3}, -\frac{1}{3}$ exact
SM average	3 generations	CKM angle	CP phase
$S = A/4G_N$ exact (Higgs)	$c_1(\mathbb{CP}^2) = 3$ exact	$\lambda_W = \tau \sqrt{G_{\text{Fermi}}}$ obs: 0.125 ± 0.001	$\delta = \arctan \varphi^2$ obs: 60.2° ± 0.05%
Mass hierarchy	Exoticity φ^2	Spectral index	QFT scale
$1 : \tau^2 : \tau^4$ top-bottom-up	33.7 vs 686 3 param. vs 0.0284	$n_s = 0.964$ obs: 0.963 ± 0.001	$M_c = 1.44 \times 10^{17} \text{ GeV}$ from μ_{QCD}/T

UNDEBATABLE PREDICTIONS

Sum of neutrino masses	$\Sigma m_\nu = 61 \text{ meV}$
Excluded: 1.940-2.447 MeV : 10 ready families	
Hall conductance in kagomé	$\sigma_{xy} = 3e^2/h$
Realistic: none : 100% topology	
Superconducting T_c	$T_c = 3b/(2k_B \tau_c)$

WHY K IS NOT A CHOICE

The SM vacuum is a composite quantum system:

- 3 colours $\rightarrow$ qutrit $\rightarrow P(\mathbb{C}^3) = \mathbb{CP}^2$
- 2 isospin $\rightarrow$ qubit $\rightarrow P(\mathbb{C}^2) \cong S^1$
- hypercharge $\rightarrow U(1) \rightarrow S^1$

$$K = \mathbb{CP}^2 \times S^1 \times S^1 \text{ is the only}$$

self-consistent geometry.

THE ONLY FREE-LOOKING PARAMETER — IS NOT FREE

$$4\tau = \cos(\pi\tau) \Rightarrow \tau = \frac{1}{2} \text{ (unique)}$$